TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders
Words: at least ≥ · at most ≤ · more than > · fewer than <.
Solve like an equation: undo + and −, then × and ÷.
Flip the sign when you multiply or divide both sides by a negative number.
Graph: ○ for < and >, ● for ≤ and ≥. Shade right for > and ≥, left for < and ≤.

1. What value of x makes $2x + 5 = 17$ true?

Tip: undo + with −, undo × with ÷

A. 11

B. 8.5

C. **6**

Subtract 5: $2x = 12$. Divide: $x = 6$.

2. Which describes the graph of $x \geq 2$ on a number line?

Tip: ≥ or ≤: closed circle · > or <: open circle

F. open circle at 2, shaded right

G. closed circle at 2, shaded left

H. **closed circle at 2, shaded right**

≥ includes 2 (closed); greater means right.

3. Is $x = -5$ a solution of $-x > 4$?

> **Tip:** put the number in for [[x]]

A. Yes, because 5 > 4

B. No, because $-5 < 4$

C. Yes, because $-5 > 4$

$-(-5) = 5$, and $5 > 4$ is true.

4. Chloe buys 2 muffins for $1.50 each and some juices for $2.25 each. Which equation gives the total cost, y, for x juices?

> **Tip:** [[y = mx + b]]: rate × amount + fixed cost

F. $y = 1.50x + 4.50$

G. $y = 2.25x + 3$

H. $y = 3x + 2.25$

Muffins: $2 \cdot 1.50 = 3$ (start). Juices: $2.25 each (rate). Like test Q6.